

9.1 Current & Potential Difference

Question Paper

Course	CIEA Level Physics
Section	9. Electricity
Topic	9.1 Current & Potential Difference
Difficulty	Hard

Time allowed: 20

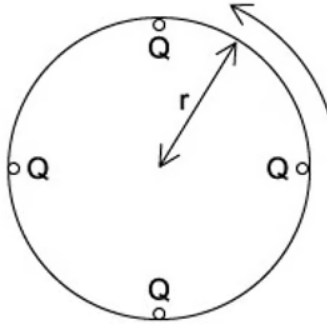
Score: /10

Percentage: /100

Question 1

Four point charges, each of charge Q , are placed on the edge of an insulating disc of radius r .

The frequency of rotation of the disc is f .



What is the equivalent electric current at the edge of the disc?

- A. $4Qf$
- B. $\frac{4Q}{f}$
- C. $8\pi rQ f$
- D. $\frac{2Qf}{\pi r}$

[1 mark]

Question 2

The amount of energy transferred when 10 C of charge passes through a p. d. of 20 V is the same as the energy needed to raise a 2 kg mass through a distance x .

Take gravitational field strength to be 10 N kg^{-1} .

What is the value of x ?

- A. 0.1 m
- B. 1.0 m
- C. 10 m
- D. 100 m

[1 mark]

Question 3

The ratings of two lamps are as follows:

X	230 V, 100 W
Y	230 V, 60 W

During normal operation, which of the lamps will have larger current and resistance?

- A. $I_X > I_Y$ and $R_X > R_Y$
- B. $I_X > I_Y$ and $R_X < R_Y$
- C. $I_X < I_Y$ and $R_X > R_Y$
- D. $I_X < I_Y$ and $R_X < R_Y$

[1 mark]

Question 4

A parallel beam of protons, each with charge e , travel at uniform velocity v . This results in current I being formed.

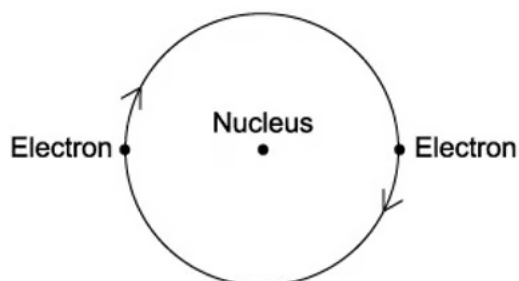
What is the number of protons present in a unit length of the beam?

- A. $\frac{I}{e}$
- B. $\frac{I}{ev}$
- C. $\frac{Iv}{e}$
- D. $\frac{I}{v}$

[1 mark]

Question 5

The diagram shows a model of an atom in which two electrons move round a circular orbit. The electrons complete 1.0×10^{15} revolutions around the nucleus every second.



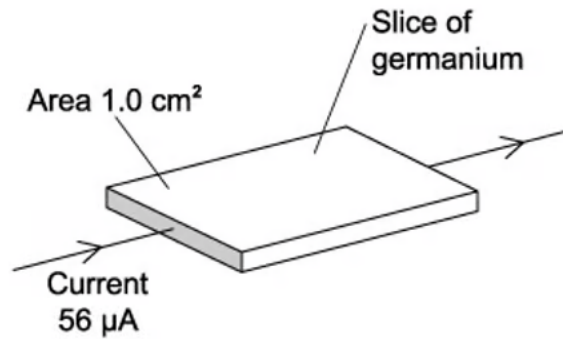
What is the current caused by the motion of the electrons in the orbit?

- A. 1.6×10^{-34} A
- B. 3.2×10^{-34} A
- C. 1.6×10^{-4} A
- D. 3.2×10^{-4} A

[1 mark]

Question 6

A slice of germanium of cross-sectional area 1.0 cm^2 carries a current of $56 \mu\text{A}$. The number density of charge carriers in the germanium is $2.0 \times 10^{13} \text{ cm}^{-3}$.



Each charge carrier has a charge equal to the charge on an electron.

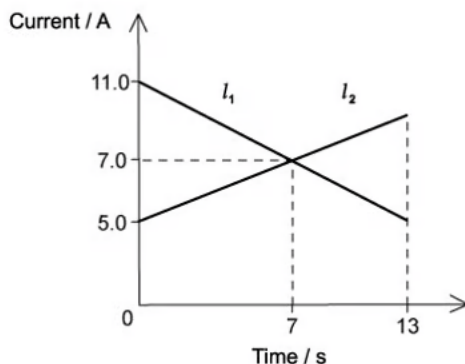
What is the average drift velocity of the charge carriers in the germanium?

- A. 0.18 m s^{-1}
- B. 18 m s^{-1}
- C. 180 m s^{-1}
- D. 1800 m s^{-1}

[1 mark]

Question 7

I_1 and I_2 are currents flowing through two different circuits, as shown in the diagram below.



Calculate the difference in the total charge that flows through the two circuits in the first 7 seconds.

- A. 0 C
- B. 21 C
- C. 42 C
- D. 49 C

[1 mark]

Question 8

The Atlantic torpedo is a large electric fish capable of generating a voltage of 220 V between its tail and its head. This drives a pulse of current of 15 A lasting for a time of 2.0 ms. The fish produces 200 pulses per second.

What is the average power output of the fish?

- A. 33 W
- B. 1.3 kW
- C. 3.3 kW
- D. 6.6 kW

[1 mark]

Question 9

A high potential is applied between the electrodes of an ionised gas. The gas carries a current of 8.16 mA and the number of electrons passing any point in the gas per unit time is $2.58 \times 10^{16} \text{ s}^{-1}$

If the charge on each positive particle is $3.2 \times 10^{-19} \text{ C}$, what is the number of positively charged particles passing any point in the gas per unit time?

- A. $1.26 \times 10^{16} \text{ s}^{-1}$
- B. $2.58 \times 10^{16} \text{ s}^{-1}$
- C. $3.84 \times 10^{16} \text{ s}^{-1}$
- D. $10.3 \times 10^{16} \text{ s}^{-1}$

[1 mark]**Question 10**

An electron beam of current 50 μA flows in a cathode-ray oscilloscope. The time-base causes the beam to sweep horizontally across the screen at $1.0 \times 10^4 \text{ cm s}^{-1}$

What is the number of electrons arriving at the screen in one centimetre length of the horizontal trace?

- A. 3.1×10^{10}
- B. 3.1×10^{13}
- C. 1.3×10^{13}
- D. 8.0×10^{14}

[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- Charge is defined as $Q = I t$
- The total charge on the disc is equal to $= Q + Q + Q + Q = 4Q$
- The frequency of rotation is equal to $f = \frac{1}{\text{period for one rotation}} = \frac{1}{t}$
 - So, $t = \frac{1}{f}$
- Current is equal to $I = \frac{Q}{t}$
- Equivalent electric current $= \frac{\text{total charge flow per rotation}}{\text{time per rotation}} = \frac{4Q}{t} = 4Qf$
 - Therefore, the equivalent electric current at the edge of the disc is $4Qf$

2

The correct answer is **C** because:

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
 - In equation form this can be expressed as $V = \frac{E}{Q}$
 - Rearranging for energy gives $E = VQ$
- So, the amount of energy transferred $= VQ = 20 \times 10 = 200J$
- Gravitational potential energy is equal to $= mgh$
 - Where h is the vertical distance moved by the mass, $h = x$
- We can equate this with the energy transferred by the charge:
 - $mgx = 200 J$
 - $x = \frac{200}{2 \times 10} = 10 m$
- Therefore, the charge is raised by 10 m

3

The correct answer is **B** because:

- Power is equal to $P = VI$
- Current is equal to $I = \frac{V}{R}$
- Combining the two equations gives
 - $P = V\left(\frac{V}{R}\right) = \frac{V^2}{R}$
- For the same value of V :
 - $P \propto I$ and $P \propto \frac{1}{R}$
- X has a higher power rating than Y, so it will have a greater current
 - $I_X > I_Y$
- X has a higher power rating than Y, so it will have a smaller resistance
 - $R_X < R_Y$

4

The correct answer is **B** because:

- The question mentions the protons travel as a parallel beam, this is a beam which does not diverge or converge – its width stays the same as the beam travels out from its source
- The equation for current in a current-carrying conductor is given by
 - $I = nAve$
 - Where n is the number density of protons and A is the cross-sectional area of the beam
- This can be written as number of protons per unit length
 - Since the unit of n is m^{-3} , the unit of A is m^2 and the unit of N is m^{-1}
- Rearranging this for N we find
 - $nA = N = \frac{1}{ev}$

5

The correct answer is **D** because:

- Charge is defined by $Q = It$
- The charge on one electron is $1.6 \times 10^{-19} \text{ C}$
- So, the total charge due to two electrons is $Q = 2 \times 1.6 \times 10^{-19}$
- The frequency of revolutions is $f = 1.0 \times 10^{15} \text{ Hz}$
- The time, or the period of one revolution, is given by $t = \frac{1}{f}$
 - So, $t = \frac{1}{1.0 \times 10^{15}} = 1.0 \times 10^{-15} \text{ s}$
- Therefore, the current is equal to $I = \frac{Q}{t} = \frac{2 \times 1.6 \times 10^{-19}}{1.0 \times 10^{-15}} = 3.2 \times 10^{-4} \text{ A}$

6

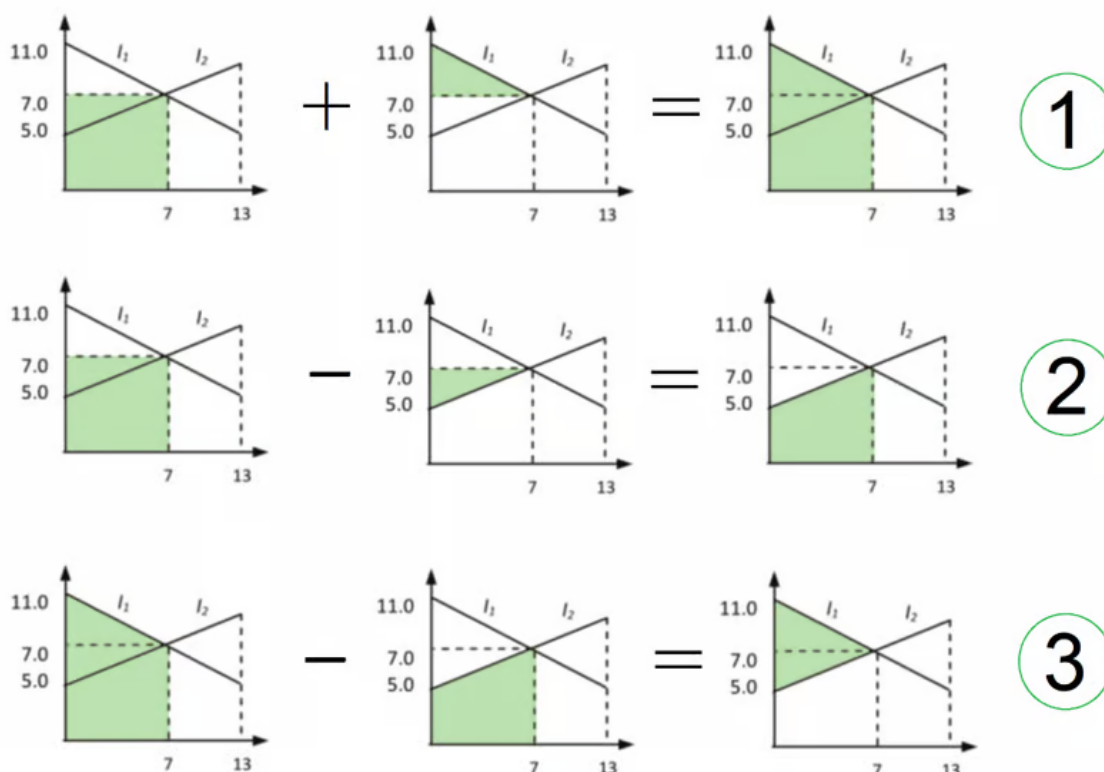
The correct answer is **A** because:

- Current in a current-carrying conductor is equal to
 - $I = nqAv$
- The drift velocity is found by rearranging for v
 - $v = \frac{I}{nqA}$
- First, we need to convert the quantities given into SI units.
- Cross-sectional area $A = 1.0 \text{ cm}^2 = 1.0 \times (10^{-2})^2 \text{ m}^2 = 1.0 \times 10^{-4} \text{ m}^2$
- Number density $n = 2.0 \times 10^{13} \text{ cm}^{-3}$
- This means, in every 1 cm^3 , there are 2.0×10^{13} charge carriers
 - $1 \text{ cm}^3 = 1 \times (10^{-2})^3 \text{ m}^3 = 1 \times 10^{-6} \text{ m}^3$
- So, in every 1 m^3 , there are $\frac{2.0 \times 10^{13}}{1 \times 10^{-6}} = 2.0 \times 10^{19} \text{ m}^{-3}$ charge carriers
- So, number density $n = 2.0 \times 10^{19} \text{ m}^{-3}$
- The charge of one electron is $q = 1.6 \times 10^{-19} \text{ C}$
 - Drift velocity $v = \frac{I}{nqA}$
 - Drift velocity $v = \frac{56 \times 10^{-6}}{(2.0 \times 10^{19})(1.0 \times 10^{-4})(1.6 \times 10^{-19})} = 0.175 \text{ m s}^{-1}$
- Therefore, the average drift velocity of the charge carriers is 0.18 m s^{-1}

7

The correct answer is **B** because:

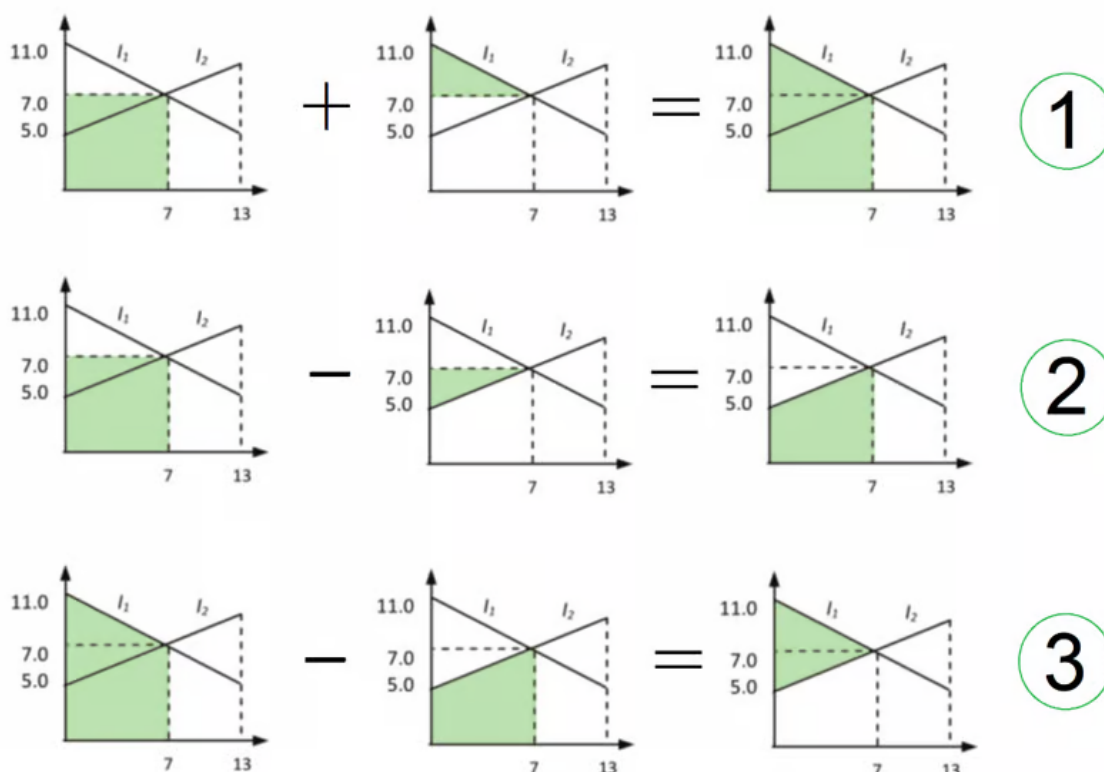
- Charge is defined by $Q = It$
- Charge can be calculated from the area under a current-time graph
- We need to find the total charge that flows through the two circuits up to 7 seconds on the graph
 - Difference in total charge that flows through the two circuits = Difference in area under graphs for the first 7 seconds
- To work this out we will need three steps (as illustrated by the diagram)
 - Step 1: Charge through $I_1 = (7 \times 7) + \frac{1}{2}(11 - 7)(7) = 63 \text{ C}$
 - Step 2: Charge through $I_2 = (7 \times 7) - \frac{1}{2}(7 - 5)(7) = 42 \text{ C}$
 - Step 3: Difference in charges = $63 - 42 = 21 \text{ C}$



Exam Tip: a quicker way to do this problem would be to just find the area shown in green in the last graph in step 3, can you see how this could be done?

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Exam Tip: a quicker way to do this problem would be to just find the area shown in green in the last graph in step 3, can you see how this could be done?

8

The correct answer is **B** because:

- Electric power is equal to $P = VI$
- Let's take another definition of power $P = \frac{E}{t}$
- We can combine these two equations to give $P = VI = \frac{E}{t}$
 - So, energy is equal to $E = VIt$
- A pulse of current of 15 A lasts for 2.0 ms
- Energy of each pulse = $220 \times 15 \times (2.0 \times 10^{-3}) = 6.6 \text{ J}$
- Average power = $\frac{\text{total energy}}{\text{time}}$
- Average power = $\text{no. of pulses} \times \frac{\text{energy of 1 pulse}}{\text{time}}$
- Average power = $\left(\frac{\text{no. of pulses}}{\text{time}} \right) \times \text{energy of 1 pulse}$
- The fish produces 200 pulses per second
 - We can write this as $\frac{\text{no. of pulses}}{\text{time}} = 200 \text{ s}^{-1}$
- Average power = $\left(\frac{\text{no. of pulses}}{\text{time}} \right) \times \text{energy of 1 pulse}$
 - Average power = $200 \times 6.6 = 1320 \text{ W}$
- Therefore, the average power output of the fish is 1.3 kW

9

The correct answer is **A** because:

- When a potential difference is applied to the ionised gas, the positive charges move towards the negative terminal and the negative charges (electrons) move towards the positive terminal
- The total current is given by the sum of these currents
 - $I = I^+ + I^-$
- Current is equal to $I = \frac{\text{total charge}}{\text{time}} = \frac{Q}{t}$
- The total charge Q is equal to the sum of charges each with charge q

- This can be expressed as $Q = Nq$
- The current of the positive charges is given by $I^+ = \left(\frac{Nq}{t}\right)^+$
 - Where $q = 3.2 \times 10^{-19} \text{ C}$
- The current of the negative charges is given by $I^- = \left(\frac{Nq}{t}\right)^-$
 - Where $q = 1.6 \times 10^{-19} \text{ C}$ and $\frac{N}{t} = 2.58 \times 10^{16} \text{ s}^{-1}$
- We can combine these equations to give
 - $I = \left(\frac{Nq}{t}\right)^+ + \left(\frac{Nq}{t}\right)^-$
 - $8.16 \times 10^{-3} = \left(\frac{N}{t}\right)(3.2 \times 10^{-19}) + (2.58 \times 10^{16})(1.6 \times 10^{-19})$
 - $\frac{N}{t} = \frac{(8.16 \times 10^{-3}) - (4.13 \times 10^{-3})}{3.2 \times 10^{-19}} = 1.26 \times 10^{16} \text{ s}^{-1}$
- Therefore, the number of positive charges passing a point is $1.26 \times 10^{16} \text{ s}^{-1}$

10

The correct answer is **A** because:

- Charge is defined by $Q = It$
- The total charge is equal to $Q = ne$
 - Where n is the number of electrons and e is the charge of one electron
- If we equate these two equations, we obtain $ne = It$
 - So, the number of electrons is given by $n = \frac{It}{e}$
- First, we must find the time taken to produce the 1 cm trace
 - $t = \frac{1}{1.0 \times 10^4} = 1.0 \times 10^{-4} \text{ s}$
- We know the current is $I = 50 \mu\text{A} = 50 \times 10^{-6} \text{ A}$
 - And the charge on an electron is $e = 1.6 \times 10^{-19} \text{ C}$
- So, the number of electrons arriving at the trace is
 - $n = \frac{It}{e} = \frac{(50 \times 10^{-6})(1.0 \times 10^{-4})}{(1.6 \times 10^{-19})} = 3.125 \times 10^{10}$
- Therefore, the number of electrons is 3.1×10^{10}